

The total possible points are 84:

P1(20): each of the linear equations in the matrix worth 2 points; being able to solve the matrix worth 3 points; stating that the matrix can be solved by successive substitution worth 3 points. Though I didn't put penalty on the process of finding the minimum of the objective function (the total cost), it's worth mentioning the right process.

Since the whole system is linear as well as the objective function, the optimum values (both maximum and minimum) for the objective function should lie at the extremes of the bounds which are Fe<sub>2</sub>O<sub>3</sub> content equal to 0 or 0.05. However, it is not clear which content value for Fe<sub>2</sub>O<sub>3</sub> is corresponding to the minimum. You need check both the values by putting them into the matrix, solving the matrix and plugging the solutions into the objective function. In the end, you can find out at Fe<sub>2</sub>O<sub>3</sub> content equal to 0.05, the objective function reaches the minimum.

P2(20): deriving the right equation for caffeine concentration worth 5 points; right solution for the time for caffeine drops to 100mg worth 3 points; plots worth 8 points; conclusion worth 4 points. Again, you can answer either it helps or not, but there should be a reason for it.

P3(14): each of the compounds worth 2 points.

P4(15): computation for the value of immunotoxin after 8 hours worth 12 points; conclusion worth 3 points. You can choose to invest by stating anticipation of the development of the technology, or not to invest for insufficient molecule accumulation.

P5(15): Stating all necessary equations worth 10 points; right solution worth 5 points.